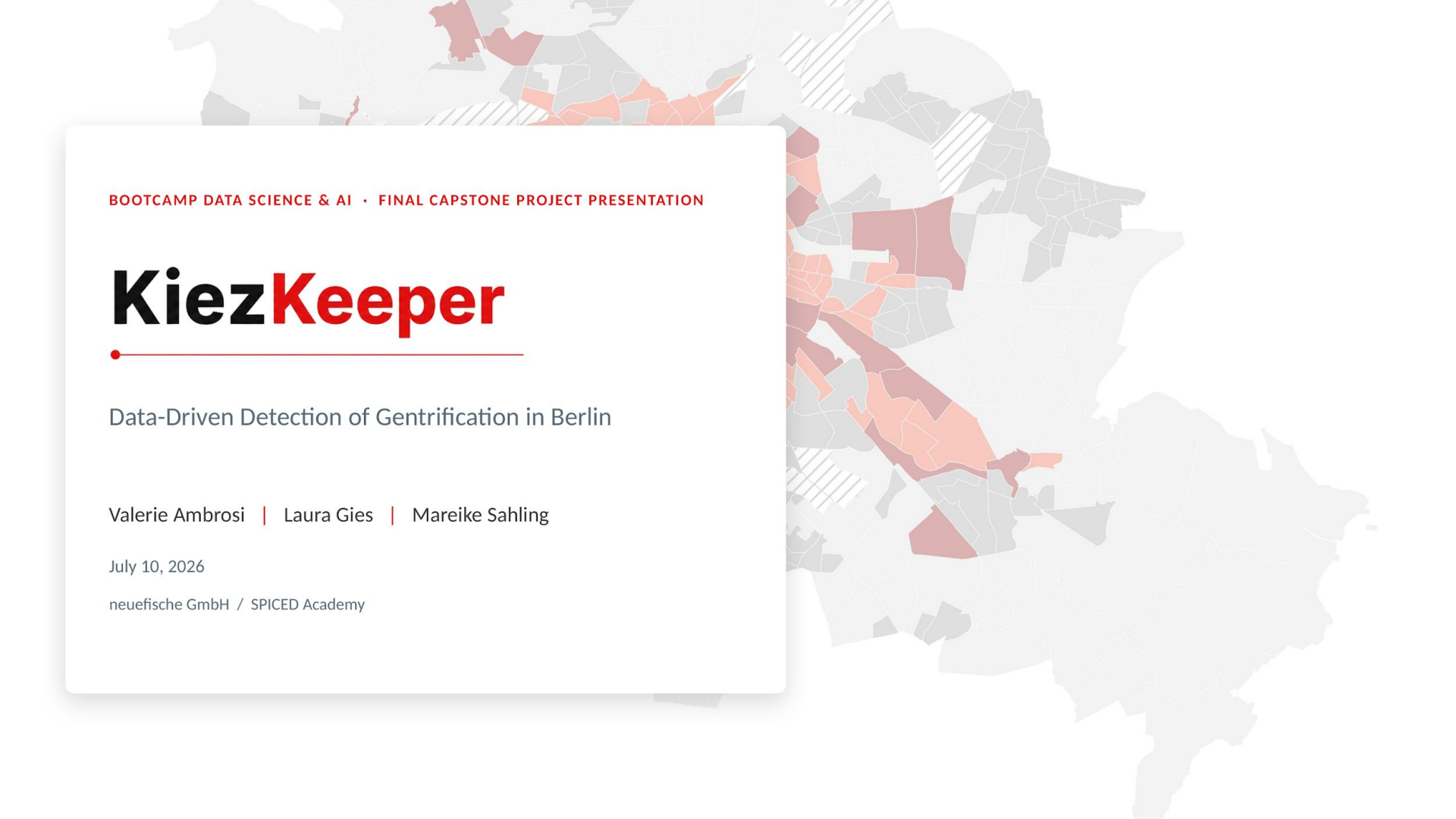




BOOTCAMP DATA SCIENCE & AI • FINAL CAPSTONE PROJECT PRESENTATION

KiezKeeper

Data-Driven Detection of Gentrification in Berlin

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July 10, 2026

neuefische GmbH / SPICED Academy

**Valerie
Ambrosi**



Bremen
Cognitive Neuroscience

**Laura
Gies**



Berlin
Political Science

**Mareike
Sahling**



Hamburg
Clinical Linguistics

**When was the last time
you had to look for an
apartment in a big city?**

Germany's housing crisis deepens as new apartment completions fall to a decade low

Brusselssignal, 22.05.2026

Greix Q1 2025: Real estate prices on the rise, especially in major cities

KIEL Institut, 08.05.2025

Berlin rents up 69%

Compared to 2016, prices have risen sharply, marking Germany's highest rent increase.

The Berliner, 08.04.2026

Rising Rents Are Putting the Squeeze on Germans

Spiegel international, 13.04.2018

Berlin: A City Plagued by Gentrification

Medium, 29.07.2019

Housing in Germany will become more expensive until 2028

Logos Press, 06.06.2026

Gentrification

Multi-dimensional neighborhood transformation

Housing market



Socio-demographic



Local economy



Gentrification

Current protection measure in Berlin:

Milieu protection (“Milieuschutzgebiete”)

Multi-dimensional neighborhood transformation

Housing market



Socio-demographic



Local economy



The Problem

Manual, inconsistent protection



Long administrative
processes



Arrives too late



Patchwork

The Problem

Manual, inconsistent protection



Long administrative
processes



Arrives too late



Patchwork

The Solution

KiezKeeper



Objective and automated



Early warning



City-wide and local

Data Extraction & Preprocessing

Real Estate

IBB Rent Reports · Zensus 2022 ·
Berlin Geoportal · Inside AirBnB

Polygon > PLR

Social

Amt für Statistik Berlin
Brandenburg · MSS

Commercial

36 monthly IHK Business
Register files (July 23 - Jun 26)

Yearly
Aggregation

Cleaning

Trend
Calculations

Final
Dataset per
PLR

Modeling: Gentrification Profiles

1: Preprocessing

KNN Imputation
Yeo-Johnson Transformer
Standardization
Block weighting

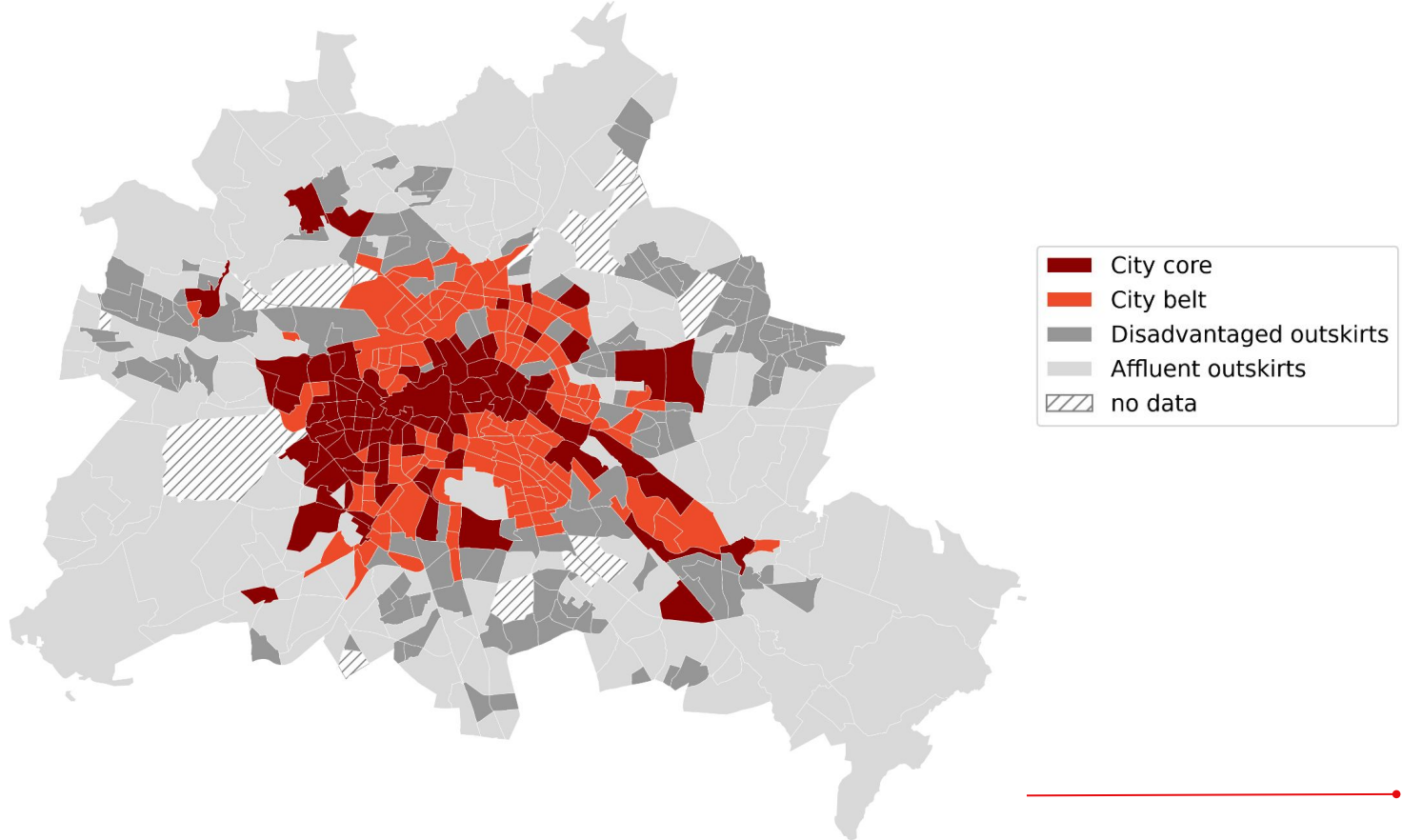
2: Clustering

Block-weighted K-means, $k = 4$

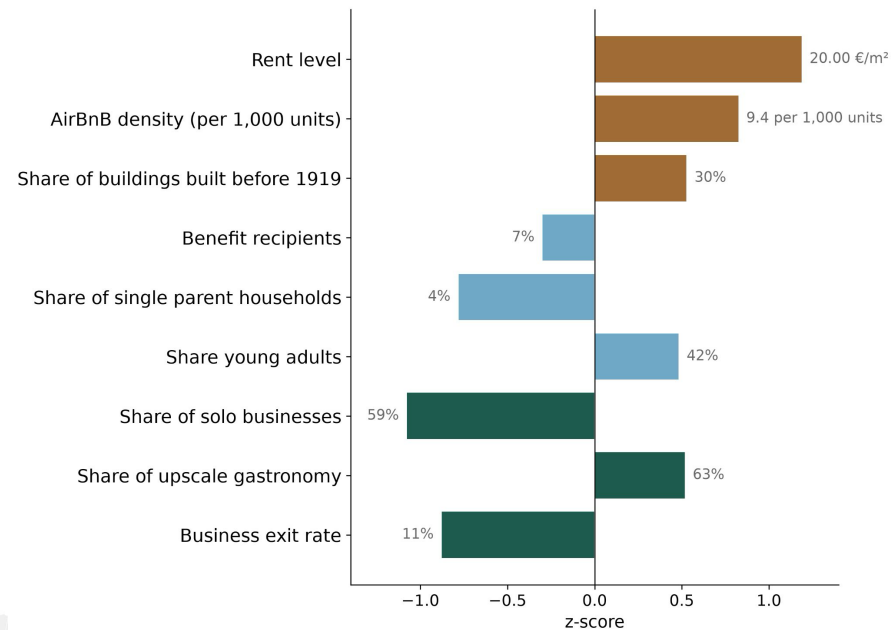
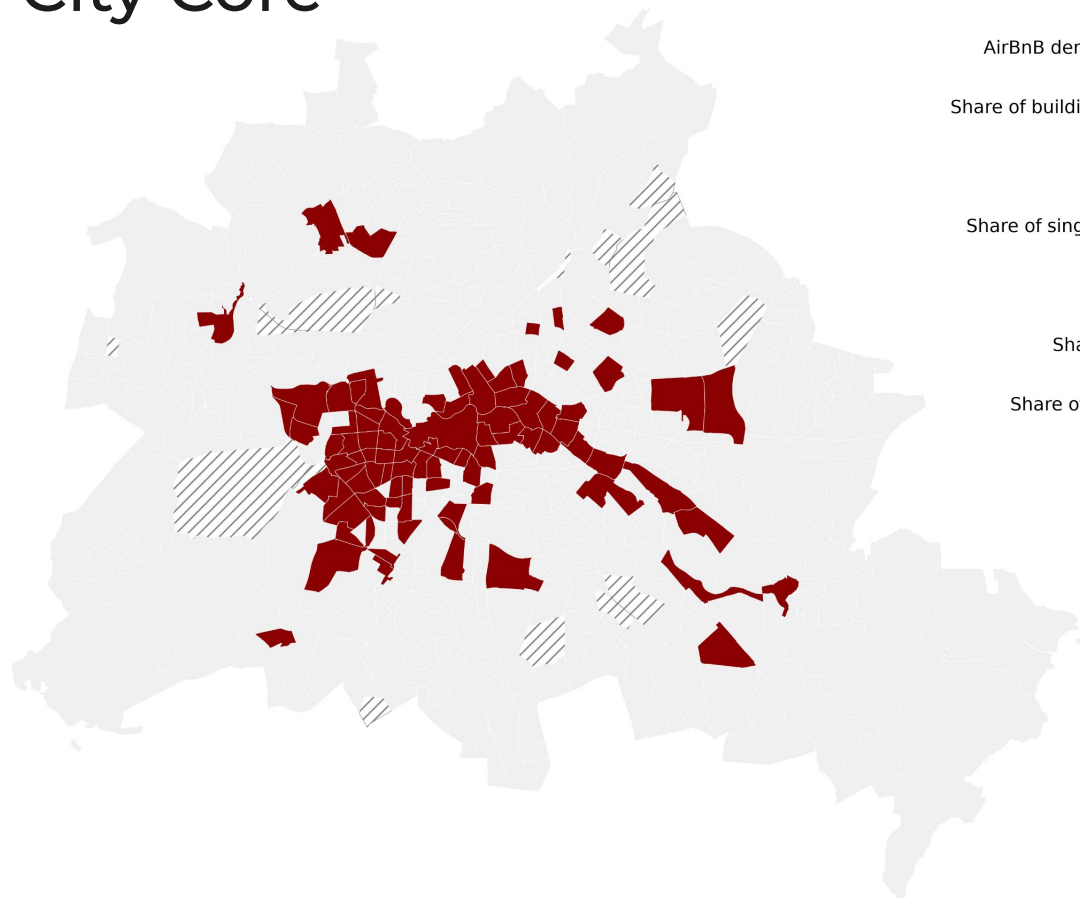
3: Validation

Stability: Subsample ARI = 0.83
XGBoost: GroupKFold ACC = 0.89

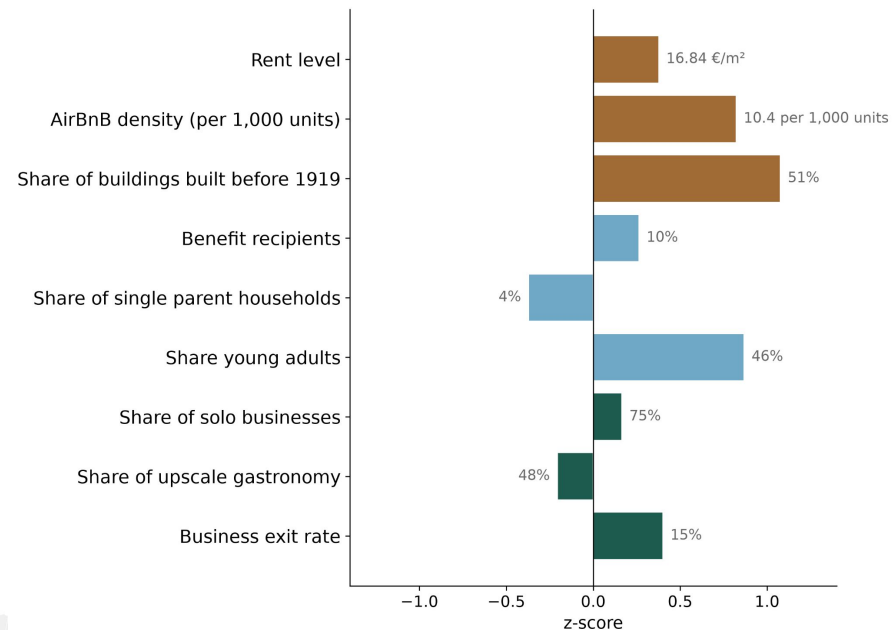
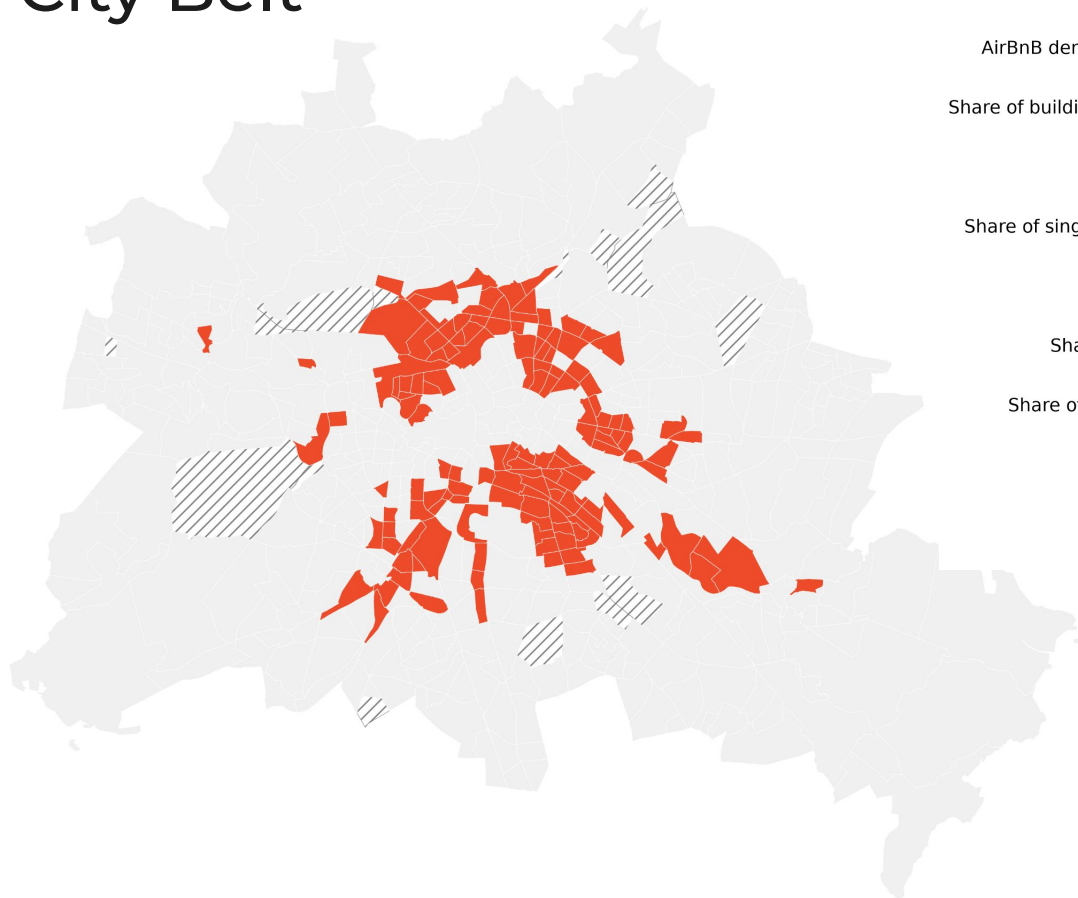
Gentrification Profiles of Berlin



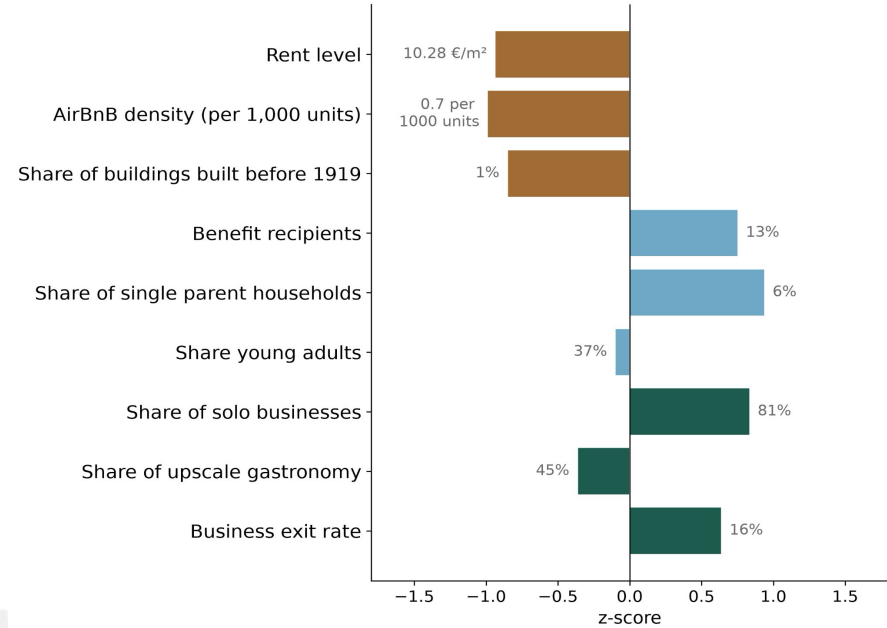
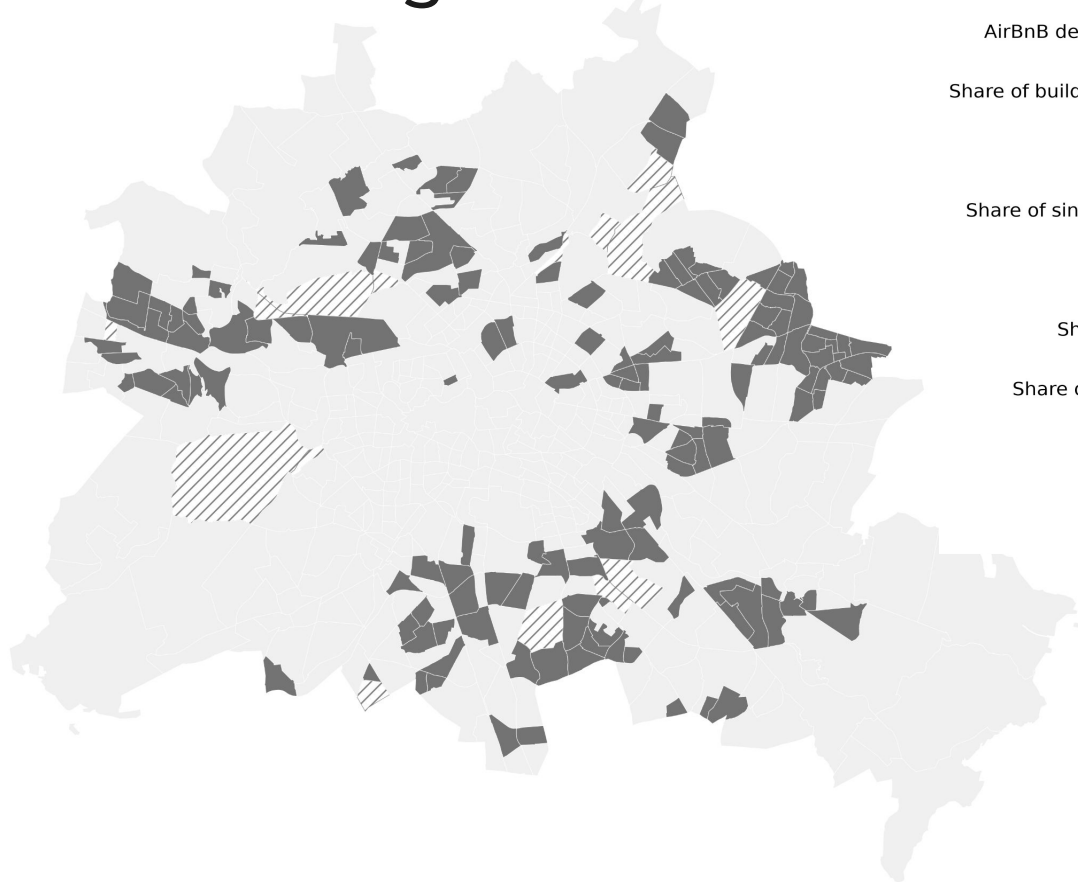
City Core



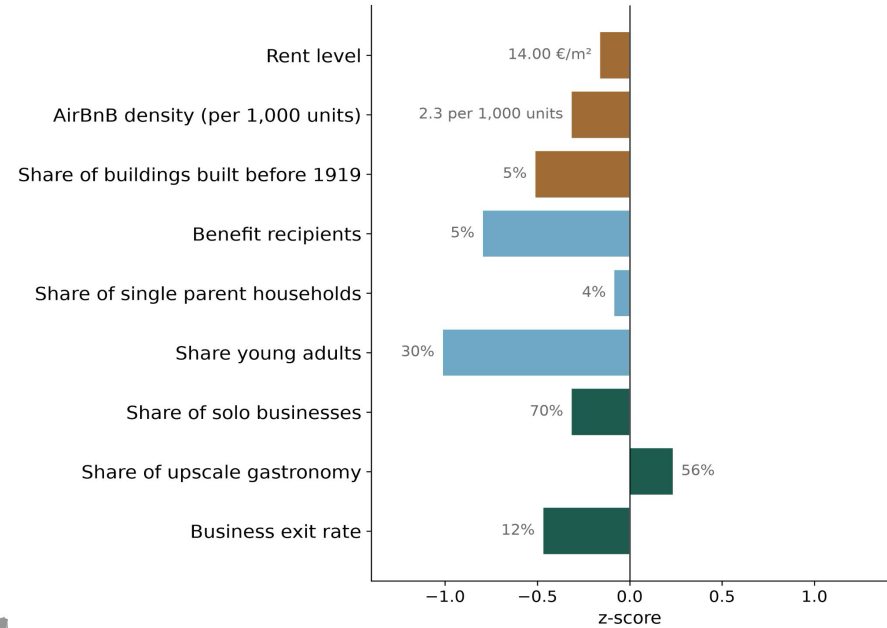
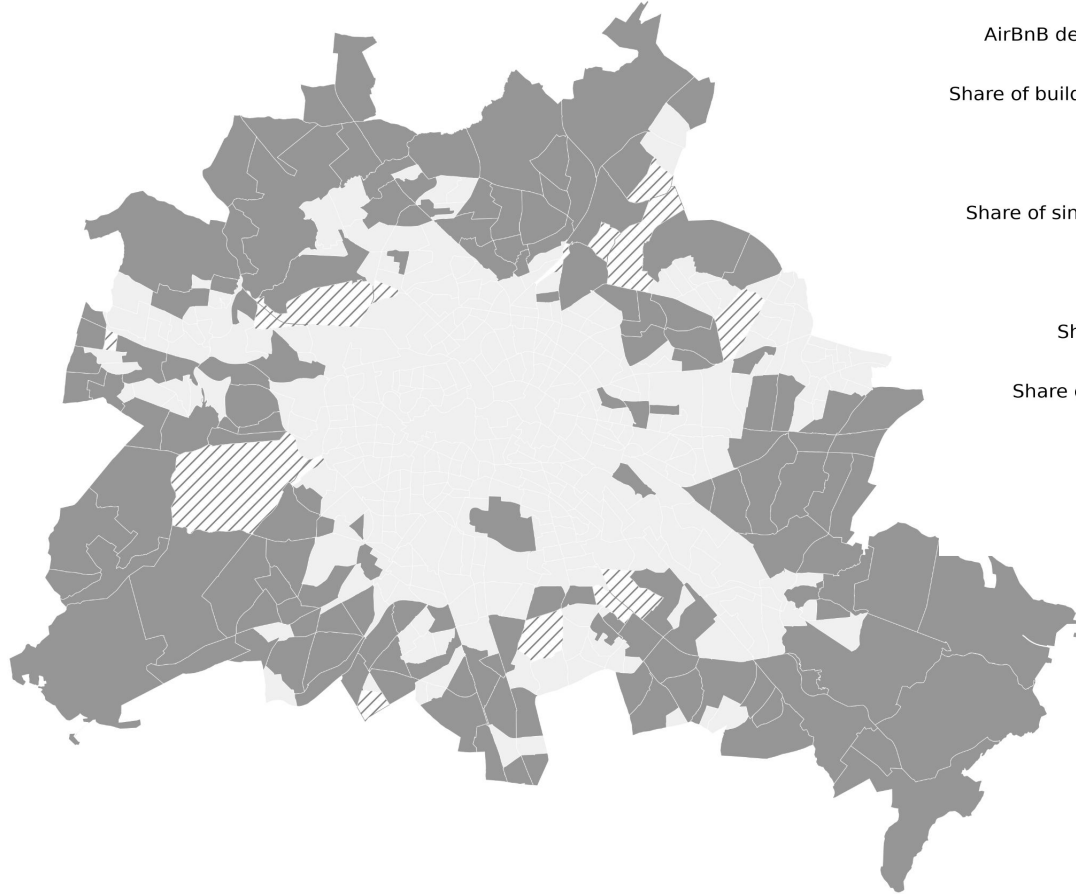
City Belt



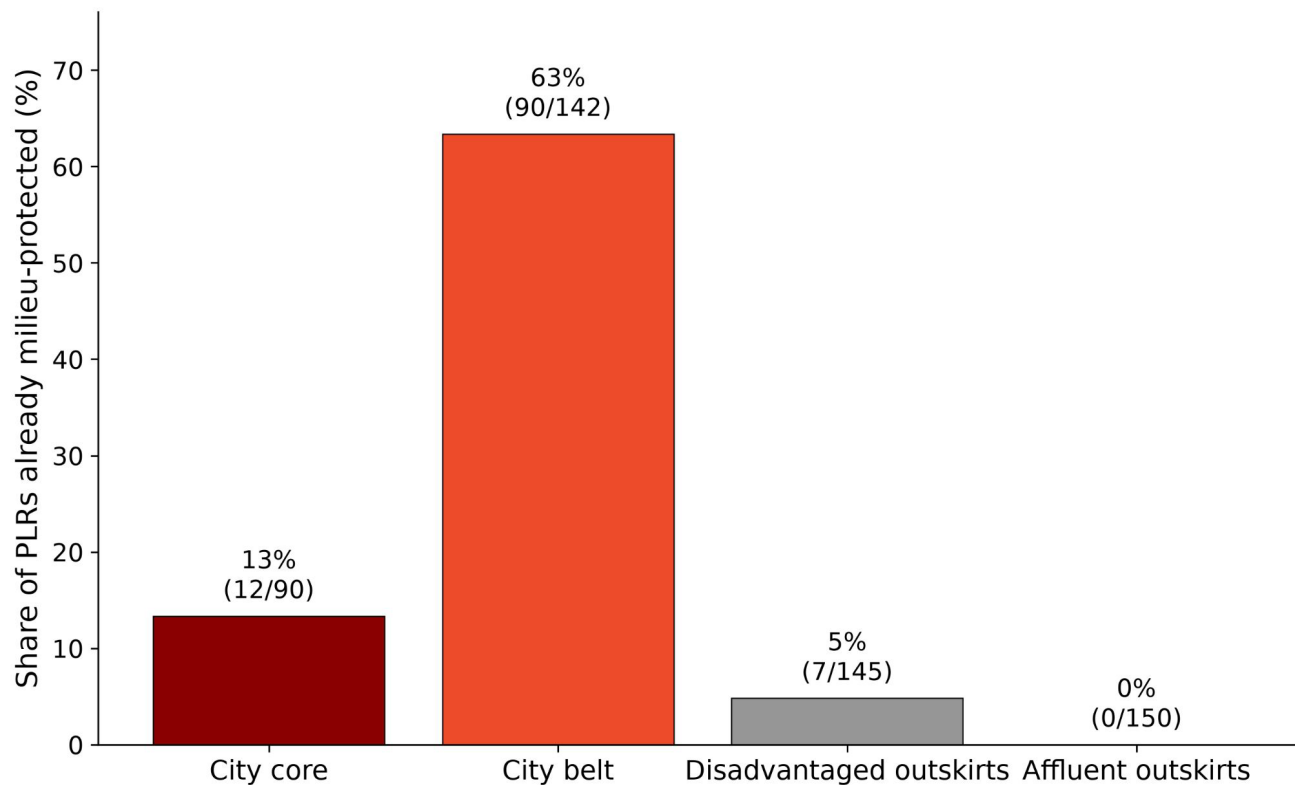
Disadvantaged Outskirts



Affluent Outskirts



Protected clusters



Modeling: Berlin's milieu protection areas

1: Preprocessing

KNN Imputation (N=5)
Milieu Protection: 0/1 and 0-1

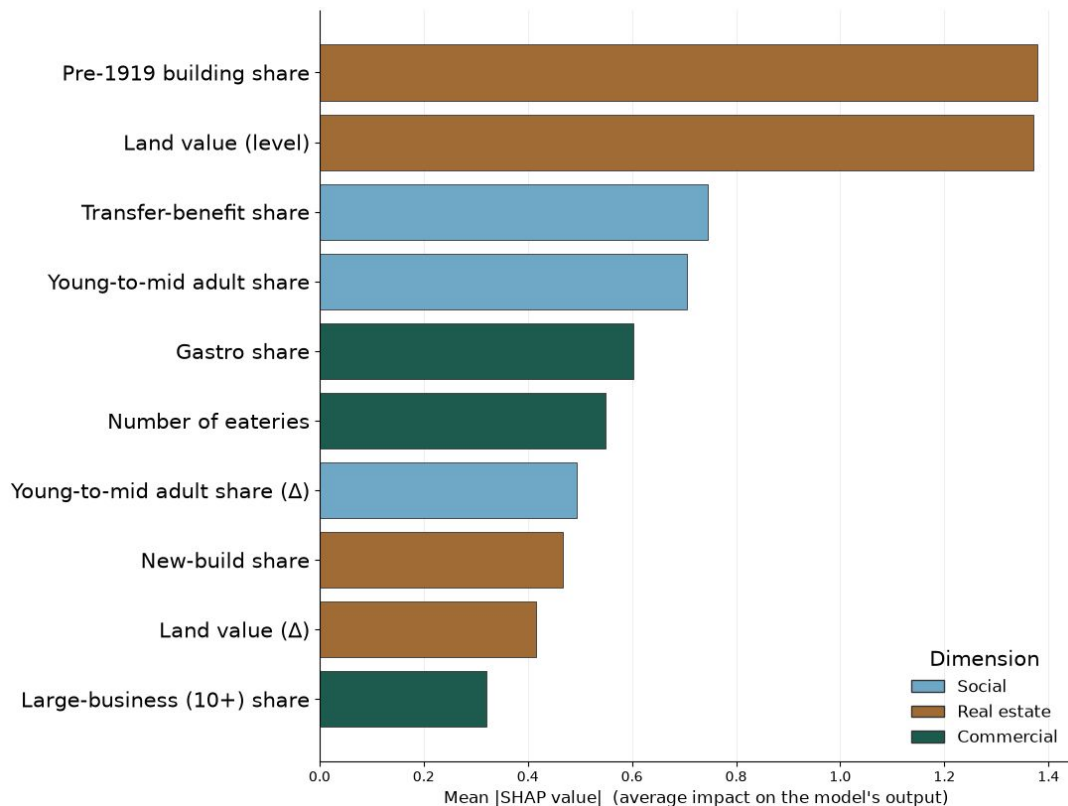
2: Modeling

OOF-GroupKFold, K=5
Protected areas: 20.7% (baseline)
Log. Regression AUC-PR: 0.63
XGBoost AUC-PR: 0.68

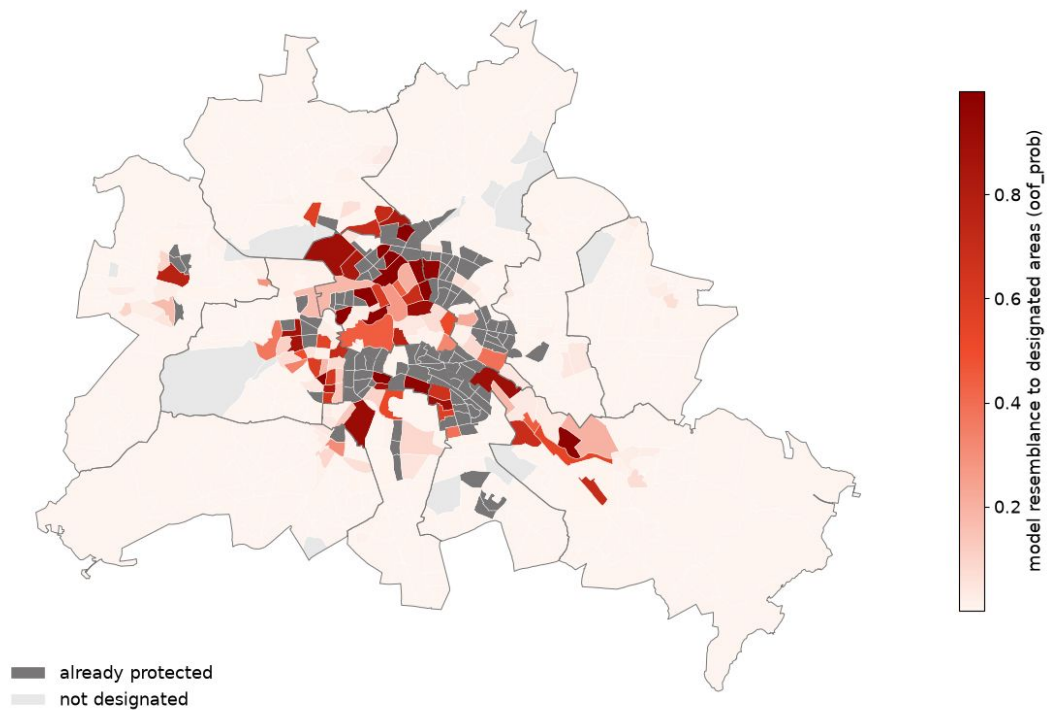
3: Validation

Cleanlab: 25/25 (+13 more)
Regression: 20/25

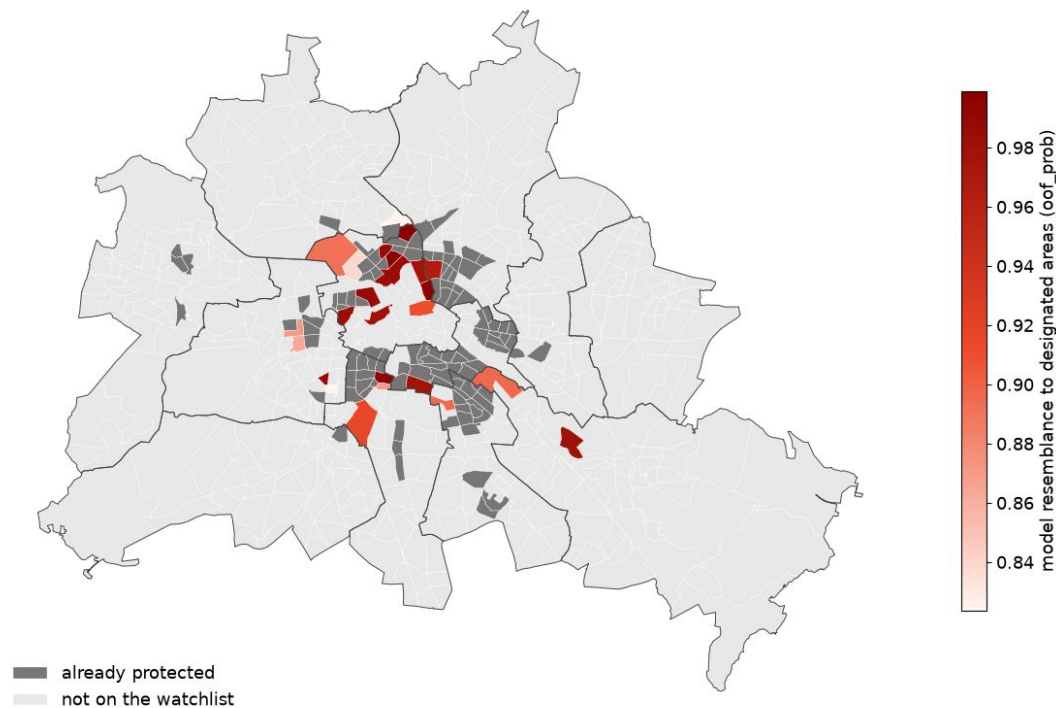
Main characteristics: Areas under protection



All areas without protection by similarity



Watchlist: the 25 with highest similarity score

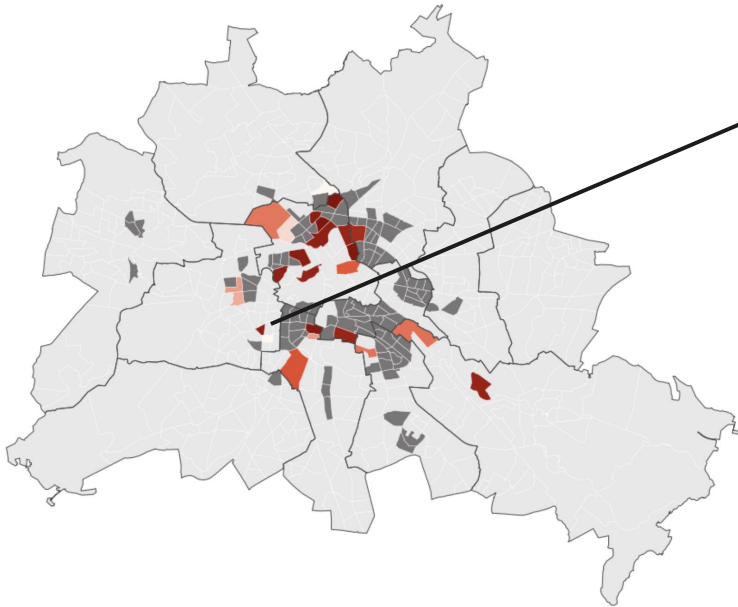


By district:

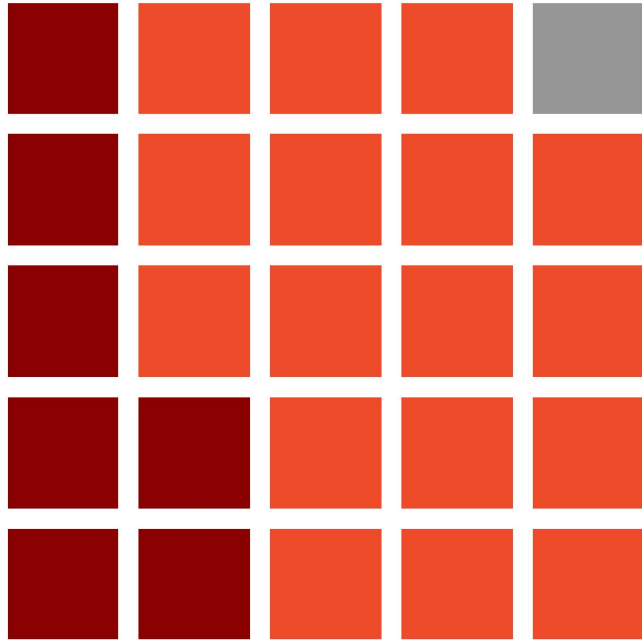
Mitte	12
Charlottenburg-Wilmersdorf	4
Tempelhof-Schöneberg	3
Treptow-Köpenick	2
Friedrichshain-Kreuzberg	1
Pankow	1
Neukölln	1
Reinickendorf	1

Breaking News!

Leon-Jessel-Platz
designated!



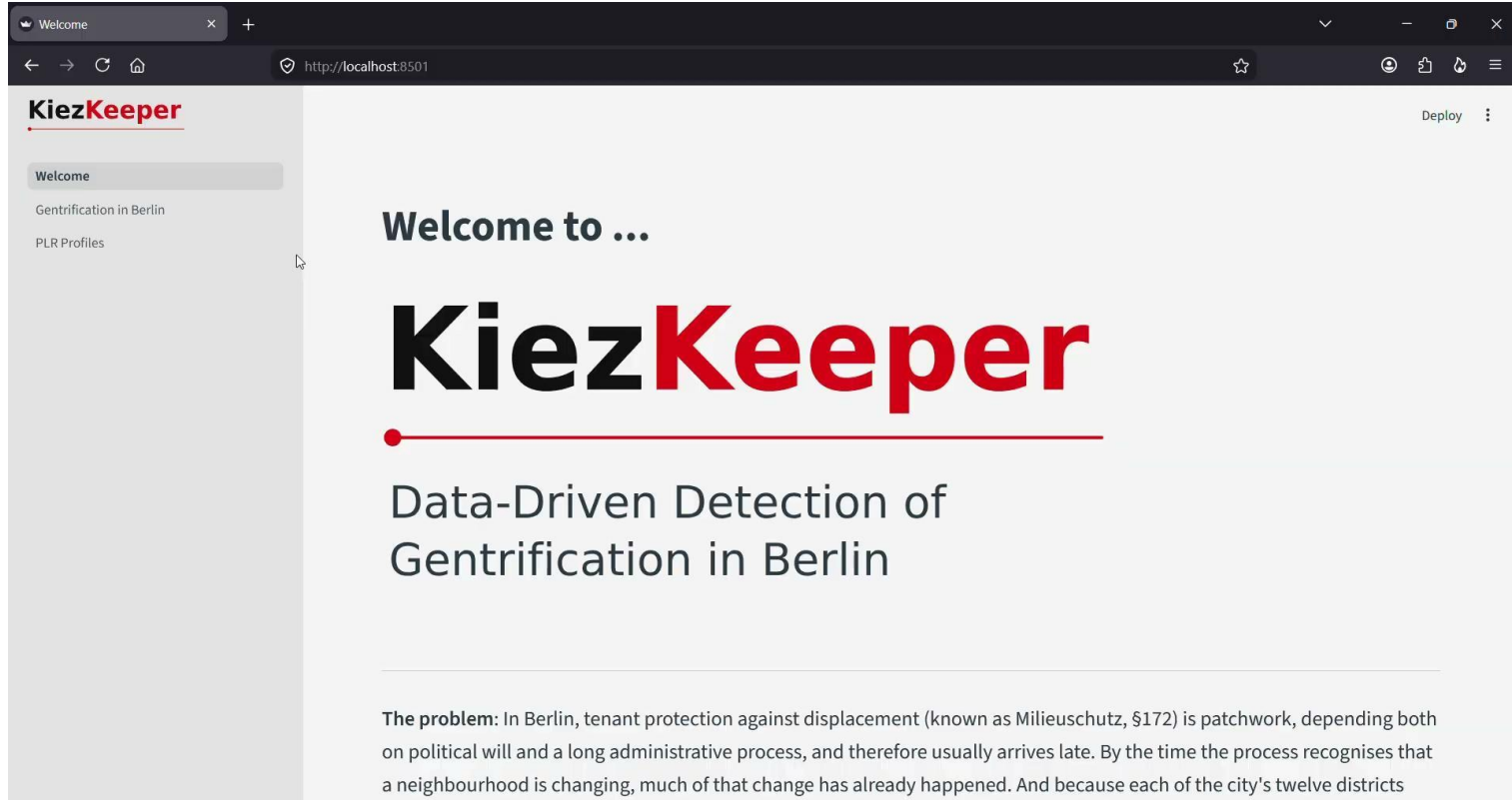
Watchlist distribution on City Profiles



- 24/25 in city core or city belt
- Two independent methods point to the same areas



Demo Video



Conclusion



City wide early warning of unprotected areas



Real-life confirmation



Include policy recommendations
See if KiezKeeper works for other cities

KiezKeeper

Thank You

Questions welcome



GitHub

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